TEMASEK JUNIOR COLLEGE

2026 PRELIMINARY EXAMINATION

Higher 2

TEMASEK JUNIOR COLLEGE

CANDIDATE NAME	
CENTRE NUMBER	S
INDEX NUMBER	

MATHEMATICS 9758/02

Paper 2 **14 September 2026**

3 hours

Additional Materials: Printed Answer Booklet
List of Formulae and Results (MF27)

READ THESE INSTRUCTIONS FIRST

Answer **all** the questions.
Write your answers on the Printed Answer Booklet. Follow the instructions on the front cover of the answer booklet.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are not allowed in a question, you must present the mathematical steps using mathematical notations and not calculator commands.
You must show all necessary working clearly.

The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **8** printed pages and **0** blank pages.

[Turn over

Section A: Pure Mathematics [40 marks]

1 A sequence $x_1, x_2, x_3, \ldots$ is defined by

$$x_1 = a \text{ and } x_{n+1} = x_n + \left(\frac{3}{2}\right)^n,$$

where a is a constant.

(a) Find x_2, x_3 and x_4 in terms of a. [2]

(b) Find x_n in the form $p + q\left(\frac{3}{2}\right)^{n-1}$, where p and q are constants in terms of a, where appropriate. [2]

(c) Determine if the sequence $\{x_n\}$ converges. [1]

2 With reference to the origin O, the points A and B have position vectors $\begin{pmatrix} 1 \\ 7 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ -2 \\ 2 \end{pmatrix}$ respectively. The point C divides the line segment AB such that $AC : CB = 2 : 1$.

(a) Find the position vector of C. [1]

The point D is the mid-point of OB and the point E has coordinates $(1, k, 0)$.

(b) Find the value of k such that C, D and E are collinear. [2]

(c) Find the possible values of k such that the length of projection of $\overrightarrow{OE}$ onto $\overrightarrow{AB}$ is $\sqrt{10}$. [3]

3 **Do not use a calculator in answering this question.**

The complex number z satisfies the equation

$$z^2(-2+\mathrm{i}) + 4\mathrm{i}z + q = 0, \text{ where } q \text{ is a real number.}$$

It is given that one root is of the form $p + p\mathrm{i}$, where p is a non-zero real number.

(a) Find the values of p and q. [4]

(b) Find the other root of the equation. [3]

4 A curve C has parametric equations

$$x = t^2 - 1, \quad y = t^3 - 3t^2 + 1, \quad \text{for } t > 0.$$

(a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of t, and hence find the coordinates of the stationary point of C. [4]

(b) Sketch the graph of C for $t > 0$. You do not need to find the intersections of the curve with the axes. [2]

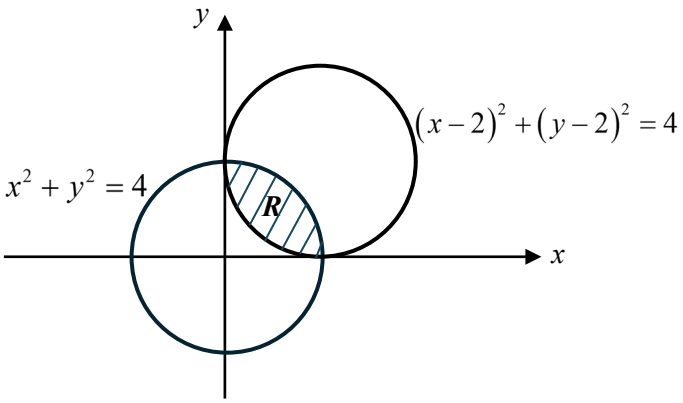
(c) Find the equation of the tangent to curve C at the point P where $t = 4$. [2]

(d) Using a non-graphical method, determine if the tangent to curve C at point P will meet C again. [3]

5 **(a)** Use the substitution $x = 2 + 2\sin\theta$ to find the exact value of $\int_0^2 \sqrt{4-(x-2)^2}\,\mathrm{d}x$. [6]

(b) The diagram below shows the two circles with equations $x^2 + y^2 = 4$ and $(x-2)^2 + (y-2)^2 = 4$. The shaded region R is bounded by the two curves. Find the exact volume of the solid formed when R is rotated through 2π radians about the x-axis. [5]

[The volume of a sphere with radius r is given by $\frac{4}{3}\pi r^3$.]



Section B: Probability and Statistics [60 marks]

6 Anand plays a game at a funfair. In each game, he is given 3 tokens which he tosses one at a time at a target area filled with red cups and black cups. It may be assumed that the probability of him tossing a token into a red cup is a constant 0.7.

Points are given for each game based on the number of tokens that are tossed into red cups. The points system is shown in the table below.

Number of tokens in red cups	Score
0	0 points
1 or 2	1 point
3	3 points

The number of points that Anand scores in one game is denoted by S.

(a) Show that $\mathrm{E}(S) = 1.659$. [3]

(b) Find $\mathrm{Var}(S)$. [1]

Anand wants to win a stuff toy that needs 100 points for redemption. He plays n games and is allowed to accumulate his points over multiple games.

(c) State the least number of games he has to play to have a chance of winning the stuff toy. [1]

(d) Find the smallest integer n for which there is at least 90% chance that Anand can redeem that stuff toy after playing n games. [4]

7 Beng, Charlie and 6 other friends bought tickets to their favourite band's concert. These tickets are for 8 consecutive seats in a row.

(a) Find the number of ways the 8 friends can seat themselves if Beng is seated next to Charlie on Charlie's left. [1]

To support their band, everyone agrees to dress in either blue or silver outfits.

(b) Find the number of ways the 8 friends can choose the colour of their outfits if there must be at least two outfits of each colour. [3]

After the concert, the 8 friends go to a café for dinner. The café has only 2 available tables: a round table with 5 seats and a round table with 3 seats.

(c) Find the number of ways they can seat themselves if Beng and Charlie are seated at the same table. [3]

8 In this question, you should state the parameters of any distributions you use.

In a game, a contestant must complete two tasks A and B in the shortest time possible. The times taken to complete task A have the distribution N(m, 1.25) and the times taken to complete task B have the distribution N(7, 1). The unit for time is minutes, and the times taken to complete the tasks are independent.

It is given that 3.1% of the contestants will take at least 22.8 minutes to complete both tasks.

(a) Show that $m = 13.0$, correct to 1 decimal place. [3]

(b) To win a prize, a contestant must take no longer than k mins to complete both tasks. The game master only wants to give a prize to the fastest 10% of the contestants who complete both tasks. Find the value of k. [2]

(c) Find the probability that the total time taken by 3 randomly chosen contestants to complete both tasks is less than 62 minutes. [3]

(d) Find the probability that the time taken by a randomly chosen contestant to complete task A is greater than twice the time taken by another contestant to complete task B. [3]

[Turn over

9 A production manager compiles data on rice crop yield, y tons/ha, at different amounts of rainfall, x cm. The data is tabulated as shown below.

x	60	80	100	120	140	160	180	200
y	2.1	2.8	3.5	4.2	4.6	4.9	5.0	4.8

The value of the product moment correlation coefficient is 0.93244 correct to 5 significant figures.

(a) What does the correlation coefficient suggest about the relationship between rice crop yield and amount of rainfall? Justify your answer. [1]

(b) Sketch a scatter diagram of the data and explain why the answer in part **(a)** should be changed. [2]

(c) Suggest a reason, in context, why a linear model $y = a + bx$, where a and b are constants, is not an appropriate model for the relationship between x and y. [1]

The production manager thinks that the data should be modelled by the quadratic equation $y = c + d(x - H)^2$, where H is a suitable constant. The product moment correlation coefficient between $(x - H)^2$ and y is denoted by r. The following table gives the values of r for some possible values of H.

H	182	183	184	185
r	–0.997383	–0.997390		–0.997293

(d) Calculate the value of r for $H = 184$, giving your answer correct to 6 decimal places. [1]

(e) Use the table and your answer to part **(d)** to suggest with a reason which of 182, 183, 184, 185 is the most appropriate value for H. Using this value for H, calculate the values of the constants c and d. [2]

(f) Give an interpretation, in context, of the values of H and c. [2]

10 In a large library, past records show that

- 80% of books borrowed are Fiction,
- 15% are Non-fiction, and
- 5% are Reference books.

Assume that each borrower can only borrow one book at a time, and each borrower borrowed books independently of one other.

(a) Three borrowers are randomly selected.

(i) Explain what is meant by randomly selected, in the context of the question. [1]

(ii) Find the probability that all three borrowed books of different types. [1]

(iii) Find the probability that at least one borrowed a Reference book. [2]

(iv) Given that at least one borrowed a Reference book, find the probability that none borrowed Fiction books. [3]

(b) A librarian selects a random sample of 10 borrowers.

(i) State, in context, one assumption needed for the number of borrowers in the sample who borrowed Fiction books to be well modelled by a binomial distribution. [1]

Assume now that the number of borrowers who borrowed Fiction books in the sample follows a binomial distribution.

(ii) Find the probability that the number of borrowers who borrowed Fiction books is between 5 and 9 inclusive. [2]

(iii) The librarian repeatedly selects 50 samples of 10 borrowers. Find the expected number of samples containing between 5 and 9 borrowers inclusive who borrowed Fiction books. [1]

(c) Find the least number of borrowers that the librarian must select in his sample such that he is at least 90% confident that there is more than one borrower in the sample who borrowed a Reference book. [3]

[Turn over

11 A tea manufacturer claims that its teabags have a mean mass of 3.2 grams. A quality control officer takes a random sample of 50 teabags and measures their masses.

The masses, x grams, are summarised as follows:

$$n = 50 \qquad \sum x = 156 \qquad \sum (x - \bar{x})^2 = 7$$

(a) Find unbiased estimates of the population mean and variance of the masses of the teabags. [2]

(b) The quality control officer wishes to test, at the 5% level of significance, whether the mean mass of the teabags has differed from 3.2 grams.

(i) State the hypotheses and find the critical region for this test. You should define any parameters you use. [3]

(ii) State the conclusion of the test in the context of the question. [2]

(c) The officer's supervisor suspects that the mean mass of the teabags is less than 3.2 grams. Given that the supervisor concluded that his suspicion is justified at $\alpha\%$ level of significance, find the range of the possible values of α. [3]